



UEFI: Should Linux Users be Worried?

Microsoft intends to capture UEFI and make GNU/Linux and other free OSs 'unauthorised'! Is this true, and if so, what is the community doing about it? This article attempts to answer some questions.

As technology enthusiasts, we are familiar with the workings of the boot mechanism in computers. We know which option to tweak in the boot settings to ensure that the CD/DVD disk is the first boot device, rather than the hard disk. Some of us might be interested in tweaking the BIOS, while still others may even have 'flashed' it at some point of time. Nothing new here, to be honest. While our desktops have come to the wobbly windows of Compiz Fusion, the BIOS still looks mundane. Yet, it does what it's meant to do and does it well for everyone—freedom lovers and proprietary software users alike.

However, in all likelihood, the days of *happy booting* as we know it may soon be gone! Yes, our 'friends' at Microsoft have done it again. They are taking one of the core aspects of computing and making it, well, painful for freedom lovers. In other words, meet UEFI, short for United Extensible Firmware Interface.

So, what's it all about?

Microsoft claims that UEFI "...brings the BIOS into the 21st century." And how does it manage to do that? Well, for a start, unlike the old BIOS, UEFI boots up instantly. It is capable of handling disks as large as over two terabytes (yes, you read that right: >2 TB). Furthermore, UEFI is independent of CPU architecture. And lastly, it brings the much needed eye-candy element to the booting mechanism. Figure 1 shows where in the stack this comes in.

Sounds good so far, doesn't it? Now, here is the tough part. Any new technology with scope for restrictions is bound to have proprietary giants lurking around it—and UEFI is no exception. To quote Gary Richmond, "... mooted changes to the UEFI firmware specification contain the implicit possibility that GNU/Linux would effectively be an 'unauthorised' operating system..."

In short, machines with UEFI on board may simply be incapable of booting GNU/Linux, BSD or any other open source OS.

But we can always over-ride the default settings, right?

Actually, no. Any changes to the UEFI firmware will require

a digitally signed image or 'key' that can only be had via the OEMs or, obviously, Microsoft. Naturally, the latter will not only deny licensing such keys, but compel OEMs to withhold them as well.

And of course, Microsoft will not admit it that easily, will they?

They are justifying this act by claiming it to be an enhanced level of protection against root-kits, boot-time viruses and other malware. Picture this: we know GNU/Linux is not malware. But what if general users of the Windows desktop wish to give Ubuntu or Fedora a spin? They can't boot it, as the UEFI mechanism will reject the boot disk as unauthorised (or, in other words, 'malware')! How is GNU/Linux going to win converts in such a case? Now, that is a tricky question!

Let's assume for a moment that Microsoft gives in and allows dual-booting GNU/Linux with Windows 8. Yet again, there is a technical catch. In order to boot under the digitally signed keys, open source boot loaders such as GRUB will need to incorporate proprietary signatures within their code. This goes against the very ideology of FOSS, and, more precisely, will not be possible under the terms of the GPL.

Okay, what now?

Before you panic, here is another angle: the digital-signature mechanism in UEFI may have a loophole that has been spotted by the community pretty recently. UEFI, generally speaking, will be supported on Windows 8 devices only (even though certain Windows versions like Vista SP1 x64 do support UEFI, the 'proper' integration will be shipped with Win8 only). As a result, older versions of Windows, such as XP, Vista, and perhaps even 7, might fall in the same category as GNU/Linux—unauthorised operating systems. While this is mere speculation, chances are that Microsoft will use UEFI as a tool to drag more users into its infamous 'upgrade cycle'—either *upgrade to Windows 8*, or, well, *upgrade to Windows 8!*